

THE ROLE OF PSEUDOPARTICLE CONFIGURATIONS IN THE SCHWINGER MODEL

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We analyze the Schwinger model in the framework of the (Euclidean) functional integral stressing the relation between pseudoparticle configurations and the structure of the model. The properties of the zero-energy modes of the Dirac equation in the presence of a pseudoparticle are discussed as well as the dynamical mass generation.

Let us consider the Lagrangian for the Schwinger model (S.M.) [1]:

$$L_S = \frac{1}{2} i \bar{\psi} \not{\partial} \psi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - g \bar{\psi} A \psi. \quad (1)$$

The field's dimensions in terms of mass are $[\psi] = [m^{1/2}]$, $[A] = [m^0]$ and for the coupling constant $[g] = [m]$. It is useful to introduce a coupling constant g_0 , adimensional, defined by the relation $g = g_0 M$, where M is an arbitrary parameter which carries the mass dimensions.

If we define $\tilde{A}_\mu = g_0 A_\mu$ and $\tilde{\psi} = g_0 \psi$, the Lagrangian reads

$$L_S = \frac{1}{g_0^2} L_S(\tilde{\psi}, \tilde{A}_\mu; M, g_0 = 1).$$

We will work then with $g_0 = 1$ and drop the tilde for simplicity. We shall analyze classical pseudoparticle contributions to the functional integral for the S.M. and which role they play. Of course, the known vacuum structure [2] of the model and the presence of a dynamical Higgs mechanism, mediated by the spinorial current, has led us to search for vortex-line configurations in Euclidean space. For that, it is necessary to have a mass term for A_μ and a suitable topological current, as is well-known from the analysis of Nielsen-Olesen [3] for the Higgs model.

Therefore, we make the ansatz that configurations of interest are solutions of the

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field equations (in Euclidean space) obtained from a massive vector field effective Lagrangian *

$$L_{\text{eff}} = L_S(\psi, A_\mu; M, 1) - \frac{1}{4}\mu^2 A_i A^i, \quad (2)$$

$$\nabla^2 A_i + \mu^2 A_i = M J_i, \quad (3a)$$

$$\partial_i A^i = 0,$$

$$(\not{\partial} - M \not{A}) \psi = D\psi = 0. \quad (3b)$$

Let us clarify this point: we introduce the effective Lagrangian (eq. (2)) in order to obtain field configurations with non-trivial topology; that is, pseudoparticles. Such configurations, as well as the solution of the Dirac equation in the presence of pseudoparticles *will reproduce the structure of the S.M.* in the path-integral framework.

The fact that the dominant contribution in the Euclidean functional integral should not come from strict minima of the action has been extensively discussed by Callan et al. [4] in the context of a Yang-Mills theory. These authors have shown that merons, as well as multiple-instanton configurations, which are not solutions of the equations of motion of the theory, are the most important ones in the discussion of quark confinement.

In our case, the relevant configuration to the S.M. will be solutions of eqs. (3a) and (3b), which are obtained from Lagrangian (2) and not from L_S . The following analysis will show how this works.

To begin with, we have in mind that the suitably defined spinorial current $J_i = \bar{\psi} \gamma_i \psi$ originates both a mass term and a vortex-type current, so that topological configurations for A_μ are induced in the functional integral; on the same ground classical solutions of the Dirac equation in the presence of those configurations play an important role.

Let us construct the “pseudoparticle” which is a solution of eq. (3a). We search for solutions of the form:

$$A_i = \epsilon_{ij} \frac{x_j}{r} f(r). \quad (4)$$

With this ansatz (consistent with the condition $\partial_i A^i = 0$), eq. (3a) becomes

$$\epsilon_{ij} \frac{x_j}{r} \left(f'' + \frac{1}{r} f' - \frac{1}{r^2} f - \mu^2 f \right) = M J_i. \quad (5)$$

* The gamma matrices are realized (in Minkowski space) by 2×2 matrices obeying, as usual $\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2g^{\mu\nu}$ and also the relation $\gamma^\mu \gamma^5 = \epsilon^{\mu\nu} \gamma_\nu$, where $\gamma^5 = \gamma^0 \gamma^1$. In Euclidean (E) space our conventions are: $x_E^1 = x^1$, $x_E^2 = ix^0$, $g_E^{ij} = -\delta^{ij}$, $\gamma_E^1 = \gamma^1$, $\gamma_E^2 = -i\gamma^0$.

This equality is only possible if $J_i = \epsilon_{ij}(x_j/r) g(r)$. Of course

$$A_i = C \epsilon_{ij} \frac{x_j}{r} K_1(\mu r), \quad (6)$$

is a solution of eq. (5) in the region where J_i can be neglected. On the other hand, asymptotically, the physical requirement is $F_{ij} \xrightarrow{r \rightarrow \infty} 0$; this implies

$$\mu A_i \xrightarrow{r \rightarrow \infty} g^{-1}(x) \partial_i g(x), \quad (7)$$

with $g(x) \in U(1)$. In other terms, A_i defines a mapping from S_1 to $U(1) (\equiv S_1)$. This can be expressed explicitly in terms of polar coordinates:

$$g(x) = \exp(-i\nu\Theta(x)), \quad (8)$$

where ν is an integer which characterizes the different classes of mappings and $\Theta(x) = \arctg(x_2/x_1)$, i.e.

$$\mu A_i^{(\nu)} \xrightarrow{r \rightarrow \infty} -\nu \partial_i \Theta(x).$$

We shall be concerned with the $\nu = \pm 1$ pseudoparticle configurations (the $|\nu| \neq 1$ case can be treated in an analogous way). In such a case, we can try to write $A_i^{\pm 1}$ in the form

$$A_i^{\pm 1} = \pm C \epsilon_{ij} \frac{x_j}{r} \left(K_1(\mu r) - \frac{1}{\mu r} \right), \quad (9)$$

where C is an arbitrary adimensional positive constant.

This form for A_i is free from singularities at the origin, satisfies the requirements at infinity and is a solution of the field equation (3a) provided J_i is given by

$$J_i = \mp C \frac{\mu}{M} \epsilon_{ij} \frac{x_j}{r}.$$

For the axial current we find (for $\nu = 1$):

$$J_k^5 = \epsilon_{kl} J^l = C \frac{\mu}{M} \frac{x_k}{r^2} = -\frac{C\mu}{M} \partial_k \log(ar), \quad (10)$$

(a is an arbitrary constant). Therefore

$$\partial^k J_k^5 = 2\pi \frac{C\mu}{M} \delta^{(2)}(r). \quad (11)$$

This result should be expected because, on one hand the form of the current J_i is dictated by the mapping defined by eq. (7); on the other hand the topological number

$$\nu = \frac{\mu}{4\pi C} \int \epsilon_{ij} F^{ij} d^2x, \quad (12)$$

is strictly correlated with the γ_5 "quantum" anomaly [7]

$$\partial^k J_k^5 = \frac{M}{2\pi} \epsilon_{ij} F^{ij} . \quad (13)$$

At this point we make a link between the present analysis and the operator description of the S.M. [11]. For that, we require the integrated form of the anomaly built with topological arguments, eq. (11), to take the same value of the integrated "quantum anomaly" for the S.M., eq. (13). The first interesting check of the consistency of our procedure is that $\mu^2 = M^2/\pi$, i.e. the correct value for the mass of the vector field. Furthermore, $J_i \sim_{r \rightarrow \infty} (M/\pi) A_i$, which reflects the existence of an effective

Higgs phenomenon [5] in the Schwinger model. The relation (12) is independent of C . However, if we want to have the usual definition of the topological number,

$$\nu = \frac{M}{4\pi} \int \epsilon_{ij} F^{ij} d^2 x ,$$

we must take $C = 1/\pi^{1/2}$.

It is interesting to note that, although the γ_5 anomaly (eq. (13)) arises from a subtle renormalization effect in the QFT, the presence of the δ -function in the classical equation for the axial current, eq. (11), reflects the same anomalous behavior. Therefore, there is an explicit relation between the topology of the configuration given by eq. (15) and the γ_5 anomaly. This result is not accidental and it has been found also in the 4-dimensional Yang-Mills theory [6].

Now, following the analysis of Callan et al. [6] for the vacuum structure, we can introduce the $|0\rangle$ states defined by $|0\rangle = \sum \exp(in\theta) |n\rangle$ where $|n\rangle$ denotes vacuum states, defined by $T|n\rangle = |n+1\rangle$. Here $T = \exp(iG_\infty)$ with

$$G_\infty = \frac{2\pi}{M} [E(\infty) - E(-\infty)] ,$$

where $E(t) = (M/2\pi) \int A_1(x, t) dx$ (in Minkowski space).

The presence of the γ_5 anomaly suppresses the vacuum tunnelling but, for an operator $D(x)$ of chirality 2, although $\langle n|D|n\rangle = 0$, $\langle n|D^\dagger(x)D(0)|n\rangle$ does not vanish for $x^2 \rightarrow \infty$ because $\langle n|D|n+1\rangle \neq 0$, i.e. the property of cluster decomposition is not satisfied. This fact implies that the $|0\rangle$ states are the physical appropriate vacuum states [2].

Then, the functional integral takes the form

$$\langle \theta' | \exp(-Ht) | 0 \rangle \xrightarrow{t \rightarrow \infty} \delta(\theta - \theta') \int \mathcal{D} A_i |_0 \exp(-\int \mathcal{L} d^2 x) \dots , \quad (14)$$

$$\langle \theta' | \exp(-Ht) D_\nu | 0 \rangle \xrightarrow{t \rightarrow \infty} \delta(\theta - \theta') \int \mathcal{D} A_i |_v \exp(-\int \mathcal{L} d^2 x) \dots , \quad (15)$$

where D_ν indicates an operator of chirality 2ν .

Let us now study the zero-energy eigenfunctions of the Dirac equation (3b) in the presence of a pseudoparticle configuration. For this purpose we note that

$$A_i^{\dagger 1} = \mp \epsilon_{ij} \partial^j \Phi(r),$$

$$\Phi(r) = + \frac{C}{\mu} (K_0(\mu r) + \log \mu r), \quad (16)$$

and

$$(\nabla^2 + \mu^2) K_0(\mu r) = -2\pi \delta^{(2)}(r),$$

$$\nabla^2 \log \mu r = 2\pi \delta^{(2)}(r). \quad (17)$$

It is easy to verify that the solutions of eq. (3b) are given by

$$\psi_{\pm 1}(x) = \exp(\mp M \gamma_5 \Phi(r)) \chi(x), \quad (18)$$

where $D\chi(x) = 0$.

Introducing the chiral decomposition

$$\chi = \begin{pmatrix} \chi_-(x) \\ \chi_+(x) \end{pmatrix}, \quad \gamma_5 \begin{pmatrix} \chi_- \\ 0 \end{pmatrix} = - \begin{pmatrix} \chi_- \\ 0 \end{pmatrix}, \quad \gamma_5 \begin{pmatrix} 0 \\ \chi_+ \end{pmatrix} = \begin{pmatrix} 0 \\ \chi_+ \end{pmatrix},$$

the two solutions which are well behaved when $r \rightarrow \infty$ take the form

$$\psi_{+1}(x) = \exp(-M\Phi) \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad (19a)$$

$$\psi_{-1}(x) = \exp(-M\Phi) \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad (19b)$$

i.e. the two zero-energy modes of eq. (3b) for $A_i^{\dagger 1} (A_i^{-1})$ have chirality +1 (−1). It should be stressed that $\psi_{\pm}$ define an orthonormal basis for the two-component spinor ψ and that the scalar density $\bar{\psi}\psi \propto \bar{\psi}_+ \psi_- + \bar{\psi}_- \psi_+$. These facts make it possible to consider the two solutions of defined chirality $\psi_{\pm}$ together. Then, we introduce the zero-mode spinor

$$\psi_0 = a_+ \psi_+ + a_- \psi_- ,$$

$$|a_+|^2 + |a_-|^2 = 1 . \quad (20)$$

We interpret the solutions (19a)–(19b) as zero-energy modes, being normalizable with the norm $\int [\psi^\dagger \psi / (1 + x^2)] d^2x$ (the factor $1/(1 + x^2)$ comes from the stereographic mapping from S_2 to E_2 . These matters have been discussed in refs. [9,12]. The results parallel the conclusions for 4-dimensional SU(2) Yang-Mills theory [8–10].

In this context we wish to stress the importance of the relation between the topology of A_i and the number of independent regular solutions of eq. (3b) with defined chirality [10]. In fact, this is a particular case of the index theory of Atiyah-

Singer for elliptic differential operators.

The zero-mode eigenfunctions (19) have the following properties:

a) $\psi_{\pm} \xrightarrow{r \rightarrow 0} \chi_{\pm}$, i.e. $\psi_{\pm}$ behaves as a zero-mass fermion for $r \rightarrow 0$,

b) $\psi_{\pm} \xrightarrow{r \rightarrow \infty} \chi_{\pm}/\mu r$, i.e. $\psi_{\pm}$ goes to zero with respect to $\chi_{\pm}$ for $r \rightarrow \infty$,

c) $\Phi(r)$ is proportional to the difference of a boson propagator of mass μ ($\mu^2 = M^2/\pi$) and a zero-mass one. This resembles strongly the structure of the fermion propagator for the S.M. as discussed by Casher et al. [11]. Finally we observe that the charge conjugation $\mathcal{C}$, applied to the Dirac equation (3b) gives $\mathcal{C}\psi_{\pm} = \psi_{\mp}$. If we interpret ψ_{+} as a fermion, then ψ_{-} must be an antifermion; so from the normalization conditions we find $N_F = N_{\bar{F}} = 1$ and defining as usual $Q = N_F - N_{\bar{F}}$ we get $Q = 0$. Then, ψ_0 given by eq. (20) is a self-charged conjugate fermion.

We will now discuss briefly how the chiral symmetry breaking induces a dynamical mass for the fermion. In doing that we have to evaluate

$$\langle \theta' | \bar{\psi} \psi | \theta \rangle = \langle \theta' | J_{+}(x) | \theta \rangle + \langle \theta' | J_{-}(x) | \theta \rangle, \quad (21)$$

where $J_{\pm}(x) = \bar{\psi}^{\dagger} \frac{1}{2} (1 \pm \gamma_5) \psi$. From eq. (15) we have

$$\begin{aligned} \langle \theta' | \bar{\psi} \psi | \theta \rangle &= \delta(\theta - \theta') \left\{ \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \mathcal{D}A \right\}_{+1} J_{+}(x) e^{-\int \mathcal{L} d^2x} \\ &+ \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \mathcal{D}A \Big|_{-1} J_{-}(x) e^{-\int \mathcal{L} d^2x} \Big\}. \end{aligned} \quad (22)$$

Using the relation [11]

$$\mathcal{D}\bar{\psi} \mathcal{D}\psi \bar{\psi} \psi e^{-\bar{\psi} D \psi} = \bar{\psi}_0 \Gamma' \psi_0 \det' D, \quad (23)$$

(the prime indicates the omission of the zero eigenvalue ψ_0 given by eq. (20)), we find in the semi-classical approximation that

$$\langle \theta | \bar{\psi} \psi | \theta \rangle = N e^{-2M\Phi(r)}. \quad (24)$$

Here we have used as classical configurations the solutions of eqs. (3a) and (3b) which should give the relevant contribution to the functional integral, eq. (22), as indicated previously. Then, we see explicitly how a dynamical mass is originated; we also note that $\langle \bar{\psi} \psi \rangle$ depends on r through the $\exp(-2M\Phi(r))$ factor, which is due to the fact that we have used the pseudoparticle contributions.

Following the same lines we can obtain any Green function for composite fermion field operators of defined chirality. For example, in the case of the correlation function $S(x) = \langle \theta | T(S(x) S(0)) | \theta \rangle$ where $S(x) = \bar{\psi}(x) \psi(x)$, using eq. (22), the relation $\psi(x) = \exp(i\gamma_5 \Phi) \chi(x)$ and performing the functional integral for A_i , in the semi-classical approximation we get

$$S(x) = N f(x^2) \exp(-2(K_0(\mu r) + \log \mu r)), \quad (25)$$

where $f(x^2)$ can be explicitly evaluated by doing the integration over the Fermi fields, which should be reducible to an integration over the free massless $\chi(x)$ fields. Therefore, we expect that an accurate analysis will give $f(x^2) \propto G_0^2(x)$, G_0 being the free fermion propagator. We will discuss the details in a future paper. If one makes the analytic continuation of eq. (25) to the Minkowski space, one finds the complete correlation function for $S(x)$ * (see e.g., ref. [11]). *In other terms, the functional integral is saturated by the pseudoparticle contributions.*

In conclusion we see the fundamental role played by the pseudoparticle in the functional integral formulation of the S.M. We have shown explicitly how the chiral symmetry breaking is originated by the pseudoparticle contributions to the γ_5 anomaly.

Furthermore, eq. (25) makes it evident that the zero-mass field quantized with negative metric, necessary for the Schwinger mechanism, has a topological origin (note the difference with the Higgs mechanism in 4 dimensions). The absence of charged sectors, i.e. $q = N_F - N_{\bar{F}} = 0$ makes possible a similar analysis to that of ref. [11], i.e. defining the multiplicative quantum number $f = +1$ (-1) for bosons (fermions), then $f = (-1)^{N_F + N_{\bar{F}}} = 1$, that is, f is screened, a necessary ingredient for "confinement" in the massless S.M. [11].

A more complete analysis should be done in order to clarify different aspects of this investigation. In this way we can reach, hopefully, a deeper insight into the relation between the topological field configurations and the charge screening present in the S.M., and then try to make a similar analysis for 4-dimensional Yang-Mills theories (it is obvious that the link between 2- and 4-dimensional theories is the γ_5 anomaly).

Another interesting model to investigate with this approach is, of course, the massive S.M. [14], because of the crucial role of the θ vacuum for its structure.

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* This aspect has been discussed in refs. [12,13] doing first the integration over the Fermi fields. Conclusions are similar but the origin and role of the pseudoparticle are different.

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